

# COMPUTER TECHNOLOGY

Computer technology is an expanding branch of engineering, with roots in both electrical engineering and computer science. It includes the related areas of computer hardware and software; computer interfacing, programming, and networking; analog and digital electronics; and robotics. Computer technicians, technologists, and engineers work in every sector of society, in careers ranging from building and repairing computer systems to designing and installing computer networks, to designing and building prototype robots and electronic devices.

Students taking computer technology courses will learn to work safely with computer equipment, electronic circuits, and robotic devices, and will explore the Essential Skills and work habits that are important for success in computer technology. Computer technology courses prepare students for apprenticeship, further study at college or university, or entry into the workplace directly after graduation.

The list of approved emphasis areas for computer technology can be found at [www.edu.gov.on.ca/eng/curriculum/secondary/teched.html](http://www.edu.gov.on.ca/eng/curriculum/secondary/teched.html).

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

# Computer Technology, Grade 11

Workplace Preparation

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TEJ3E

This course enables students to develop knowledge and skills related to computer hardware, networks, operating systems, and other software. Students will use utility and application software, and learn proper procedures for installing, maintaining, and troubleshooting computer systems and networks. Students will develop an awareness of environmental and societal issues related to the use of computers, and will learn about apprenticeships and other employment opportunities in the field of computer technology that they may choose to pursue after graduation.

**Prerequisite:** None

# A. COMPUTER TECHNOLOGY FUNDAMENTALS

## OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the function and development of a variety of computer hardware;
- A2.** describe the topologies and hardware of computer networks, and the advantages and disadvantages of computer networking;
- A3.** describe the requirements and purpose of a variety of current software, and identify compatibility issues for this software.

## SPECIFIC EXPECTATIONS

### A1. Computer Hardware

By the end of this course, students will:

- A1.1** use relevant technical terminology to describe computer hardware (*e.g., processors, memory, drives, monitors, printers*);
- A1.2** identify and describe current storage devices (*e.g., hard drives, CD/DVD drives, USB drives*);
- A1.3** describe the function of current input and output devices (*e.g., keyboards, mice, tablets, printers, monitors, scanners, webcams*);
- A1.4** discuss past developments and current trends in hardware technology (*e.g., changes in types, capacity, speed, and cost*).

### A2. Networking Concepts

By the end of this course, students will:

- A2.1** describe common network topologies and requirements (*e.g., physical layouts, equipment, connections*);
- A2.2** describe the function of various types of network hardware (*e.g., hubs, switches, routers, cabling*);
- A2.3** describe the advantages and disadvantages of networked computing (*e.g., data sharing, collaborative applications, security, costs, centralized administration, reliability, effects of equipment failure*).

### A3. Software

By the end of this course, students will:

- A3.1** describe the purpose of various types of software (*e.g., operating systems, application software, drivers, firmware*);
- A3.2** compare the requirements of various operating systems (*e.g., memory, hard-drive space, processor type and speed, video resolution*);
- A3.3** identify a variety of current software products, their uses, and their hardware requirements;
- A3.4** describe the use of drivers and plug-and-play systems for computer hardware;
- A3.5** identify issues of software compatibility (*e.g., operating systems; software, driver, and operating system updates; hardware; backward and forward compatibility of files*).

## B. COMPUTER TECHNOLOGY SKILLS

### OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** research and determine hardware solutions for users' computer needs;
- B2.** install, update, maintain, and troubleshoot computer hardware, and back up system and user data;
- B3.** plan, install, and manage a computer network;
- B4.** install, use, and update operating systems, utility software, and application software.

### SPECIFIC EXPECTATIONS

#### B1. Hardware Solutions

By the end of this course, students will:

- B1.1** analyse user hardware needs for a personal computer (*e.g., processor type and speed, memory, hard-drive capacity, modem and/or network card, video card, audio card, number and types of ports, wireless communication*), and write a summary of these needs;
- B1.2** compare availability and costs for hardware from local, national, and/or global suppliers;
- B1.3** design an effective home office layout, including the location of computers, peripherals, furniture, and lighting.

#### B2. Installation, Maintenance, and Troubleshooting

By the end of this course, students will:

- B2.1** follow established procedures for installing, servicing, and troubleshooting computer hardware;
- B2.2** follow correct procedures to prevent damage to computer components (*e.g., use of anti-static wrist straps, mats, bags, and containers*);
- B2.3** perform basic maintenance on the hardware components of a computer system (*e.g., remove dust from fans and circuit boards, clean input devices and monitor screen, change printer cartridges*);
- B2.4** use utility software to diagnose and correct problems (*e.g., defragment a drive, scan hard drives for errors and defective sectors, detect and remove viruses and spyware*);

- B2.5** use established procedures to perform back-ups of system data and user information;

- B2.6** install and configure expansion and upgrade devices (*e.g., video card, network card, memory, secondary storage devices*).

#### B3. Network Setup and Management

By the end of this course, students will:

- B3.1** use a design process (see pp. 22–23) to plan and install a home or small office network;
- B3.2** install and configure additional shared devices on a network (*e.g., printers, scanner, drives*);
- B3.3** use network utilities (*e.g., ping, telnet, ipconfig*) to diagnose and correct problems (*e.g., incorrect IP configuration, slow connections*).

#### B4. Software Implementation

By the end of this course, students will:

- B4.1** collect and analyse information from customers to determine their software needs (*e.g., security requirements, budget, applications, preferred platform*);
- B4.2** properly set up and configure software for various user needs (*e.g., business, personal, and/or gaming applications; access for multiple users and/or visually impaired users; master user account*);
- B4.3** properly install software upgrades (*e.g., operating system, application software*);
- B4.4** install and update utility and antivirus software.

# C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

## OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe environmental issues related to the widespread use of computer technology;
- C2.** describe societal issues related to the widespread use of computer technology.

## SPECIFIC EXPECTATIONS

### C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the effects of computer technology on the environment (*e.g., accumulation of electronic waste, use of lead and other toxic materials in computers, use of ozone-destroying chemicals to wash soldering flux from circuit boards, energy consumed by computers left in standby mode, energy saved by use of programmable thermostats*);
- C1.2** outline how community partners and government agencies apply the reduce/reuse/recycle concept to computer technology.

### C2. Technology and Society

By the end of this course, students will:

- C2.1** describe the benefits of computer technology for society (*e.g., cheaper and more efficient information services, telecommuting, development of a “global village”*);
- C2.2** describe the drawbacks of computer technology for society (*e.g., Internet gambling addictions, more sedentary lifestyle, spam, telemarketing, loss of privacy*).

# D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

## OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** identify and follow computer-related safety practices;
- D2.** describe ethical and security issues related to the use of computers;
- D3.** describe and apply professional customer-service practices;
- D4.** apply the skills required for success in the workplace;
- D5.** describe opportunities for careers and training related to computer technology.

## SPECIFIC EXPECTATIONS

### D1. Health and Safety

By the end of this course, students will:

- D1.1** use appropriate equipment, techniques, and strategies to avoid health and safety problems when assembling, using, and maintaining computer systems (*e.g., repetitive strain injuries, eye strain, electrical shock*);
- D1.2** describe issues related to Internet safety (*e.g., protection of information stored on computers or transmitted over a network, cyberstalking, cyberbullying, privacy policies*).

### D2. Ethics and Security

By the end of this course, students will:

- D2.1** comply with acceptable-use policies for computers (*e.g., restrictions on commercial or personal use, prohibition of inappropriate content and plagiarism, protection of privacy and intellectual property rights*);
- D2.2** explain how copyright legislation applies to computer software and media content, and follow proper licensing and registration procedures for such material (*e.g., rights: copyright, open source, public domain; licences: shareware, retail, copyleft such as Creative Commons and GNU*);
- D2.3** explain the importance of passwords, security software updates, and protection of personal information and client data.

### D3. Customer Service

By the end of this course, students will:

- D3.1** keep accurate records of customer information using a database, spreadsheet, or word processor (*e.g., customer name, contact information, contact date and time, description of the technical problem and any action taken*);
- D3.2** describe the importance of professionalism in customer service (*e.g., telephone etiquette, appropriate personal appearance and demeanour, keeping accurate records*);
- D3.3** follow appropriate customer-service protocols and procedures (*e.g., procedures for dealing with complaints, troubleshooting, and customer support by telephone, email, or the Internet*);
- D3.4** communicate with clients using an appropriate level of technical terminology.

### D4. Workplace Skills

By the end of this course, students will:

- D4.1** use time-management skills in project settings (*e.g., set realistic goals, recognize time constraints, plan for deadlines*);
- D4.2** use computer terminology correctly, and compile an up-to-date glossary of computer terms and acronyms.

## D5. Career Opportunities

By the end of this course, students will:

**D5.1** describe various computer-related job opportunities in local business and industry (e.g., retail sales, apprenticeship, IT hardware technician, IT network technician, electronic service technician);

**D5.2** identify opportunities for further training and certification (e.g., apprenticeship, college courses, trade certifications);

**D5.3** describe lifelong learning methods used in the computer technology industry (e.g., web-based learning, printed materials, college courses, on-the-job training);

**D5.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in computer technology (e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations);

**D5.5** demonstrate an understanding of and apply the Essential Skills that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (OSP) (e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills);

**D5.6** demonstrate an understanding of and apply the work habits that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (e.g., working safely, teamwork, reliability, organization, working independently, initiative, self-advocacy, customer service);

**D5.7** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in computer technology (e.g., Passport to Safety certificate, OSP Work Plan, OSP Transition Plan, work logs, photographs of projects), and explain why having a current portfolio is important for career development and advancement.